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UNION OF THE COMOROS

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STAFF REPORT FOR THE 2023 ARTICLE IV CONSULTATION, FIRST REVIEW UNDER THE EXTENDED CREDIT FACILITY ARRANGEMENT, REQUESTS FOR WAIVERS OF NONOBSERVANCE OF PERFORMANCE CRITERIA, AND MODIFICATIONS OF PERFORMANCE CRITERIA—DEBT SUSTAINABILITY ANALYSIS

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Risk of external debt distress	*High*
Overall risk of debt distress	*High*
Granularity in the risk rating	*Sustainable*
Application of judgement	*No*

Comoros external and overall debt are sustainable including in the medium term but remain at high risk of debt distress[1]. Three of the four external debt burden indicators continue to breach their respective thresholds under the baseline. However, mild improvements in fiscal balances from higher-than-expected revenues in 2023H1 as well as the lower-than-anticipated financing needs over 2023-25 have led to improvements in the external debt burden and liquidity indicators compared to the DSA of the ECF request (May 2023). Factors that mitigate the mechanical risk signals include the authorities' reform plans supported by the IMF's Extended Credit Facility (ECF) and the World Bank's Development Policy Operations (DPOs), as well as the consistently strong performance of remittances, which supports external sustainability. This overall assessment is contingent on: (i) progress on domestic resource mobilization and gradual but sustained fiscal consolidation over the medium term, (ii) no additional non-concessional borrowing, and (iii) enhanced financial oversight of SOEs. The DSA suggests that Comoros' external debt is susceptible to export and contingent liability shocks.

[1] Comoros' debt carrying capacity is assessed as medium, given a Composite Indicator of 2.82 based on October 2023 WEO projections and the 2022 Country Policy and Institutional Assessment (CPIA) rating.

PUBLIC DEBT COVERAGE

1. Public debt coverage includes all financing from the central bank to the Treasury. The coverage of external debt includes the entire public sector, namely external debt of the central government, the central bank borrowing on behalf of the government, and government-guaranteed debt of SOEs. Subnational government entities cannot take up external debt on their own, and they have not taken on domestic debt. SOEs cannot access the external debt market without a government guarantee. Domestic debt includes central government, SOE's guaranteed debt from domestic banks, and on-lending from the Central Bank of Comoros (BCC) to the government, including statutory advances and loans connected to the IMF SDR allocation.

Text Table 1. Union of the Comoros: Public Debt Coverage

Subsectors of the public sector	Sub-sectors covered
1 Central government	X
2 State and local government	
3 Other elements in the general government	
4 o/w: Social security fund	
5 o/w: Extra budgetary funds (EBFs)	
6 Guarantees (to other entities in the public and private sector, including to SOEs)	X
7 Central bank (borrowed on behalf of the government)	X
8 Non-guaranteed SOE debt	

Sources: Comorian authorities, and IMF staff

Text Table 2. Union of the Comoros: Coverage of the Contingent Liabilities Stress Test

1	The country's coverage of public debt	The central government, central bank, government-guaranteed debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	4.5	Reflects unaudited estimates of domestic arrears.
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	3	SOEs growing debt
4	PPP	35 percent of PPP stock	0	
5	Financial market (the default value of 5 percent of GDP is the minimur	5 percent of GDP	5	
Total (2+3+4+5) (in percent of GDP)			12.5	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

Sources: Comorian authorities, and IMF staff

2. Contingent liabilities have built up significantly and the likelihood of their materialization is growing. The magnitude of the contingent liabilities shock is 12.5 percent of GDP reflecting: (i) preliminary outcomes of an ongoing audit pointing to a higher-than-expected level of domestic arrears[1] and (ii) difficulties encountered by the SOEs involved in importing food and oil in rolling over their trade loans in 2022 following the deterioration of their cash flows due to higher commodity prices (1 percent of GDP added to the default 2 percent of GDP to capture unknown cross-debts of SOEs), and (iii) the average cost to the government of a financial crisis in low-income countries since 1980—5 percent of GDP also captures about 1/3 of banking sector assets in Comoros (Text Table 2).

[1] These domestic arrears represent unpaid bills for goods and services and are included as contingent liabilities as the figures are still preliminary. It includes arrears from the treasury to the pension system. Audit of domestic arrears are ongoing. Once completed, an arrears clearance strategy will be identified.

BACKGROUND

A. Recent Debt Developments

3. Public debt ratio has been on an upward trend, rising from 30.6 percent of GDP in 2021 to 34.0 in 2022. External public debt, which constitutes around 80 percent of total public debt, reached 27.3 percent of GDP in 2022 and is expected to rise further to 32.5 percent of GDP by end-2023, up from 20.4 percent of GDP prior to the pandemic. Domestic public debt jumped from 5.2 percent of GDP in 2021 to 6.8 percent of GDP in 2022, reflecting the BCC's lending to the government in connection to the SDR allocation and a higher uptake of publicly guaranteed domestic debt by the SOEs involved in food and energy supply. Recent stabilization of commodity prices during 2023 has reduced losses at key SOEs; as such, domestic debt is expected to decline to 6 percent of GDP (vs. 7.2 percent projected in the previous DSA).[2] The authorities have made progress in public debt management with the promulgation of a debt management law which is a FY23 public policy action (PPA) of the sustainable development policy financing (SDFP), and a prior action under the World Bank's DPO approved on September 29, 2023.

4. The composition of external public debt is dominated by non-Paris Club lenders, and especially China. At end-2022, disbursed debt owed to official bilateral creditors accounted for 61 percent of public external debt, debt to multilateral creditors for 19 percent, and debt to regional lenders for 20 percent. The share of non-concessional debt stood at 10 percent of external debt at end-2022 and is expected to grow in the next few years as the non-concessional loan for the construction of the Galawa Hotel is disbursed.

5. External arrears would increase from USD7.2 million at the end-2022 to 8.2 million at end-2023 (0.6 percent of the GDP).[3] A large part of the outstanding stock is accounted for by the loan from the Arab Bank for Economic Development in Africa (BADEA), for which the authorities are seeking an arrears clearance strategy. The authorities have an arrears clearance agreement with Bpifrance Assurance Export, covering a partial amount (USD 0.18 million), while negotiations are underway for the remaining amount (USD 0.6 million). Discussions are also ongoing for the arrears to Exim Bank India which concern a disputed power plant project.

6. The authorities are mobilizing concessional loans for the construction of the El Maarouf Hospital. The latest estimated cost to complete the hospital has been lowered from USD 70 million to USD 55 million, which is included in this baseline.

[2] Domestic debt to SOEs decreased by 8 percent (q/q) in June 2023.

[3] Projections of debt service payments to clear the arrears are included in macroeconomic assumptions underlying this exercise. The arrears represent an amount lower than 1 of GDP and hence do not trigger an in-debt distress rating on a de minimis basis.

B. Macroeconomic Forecasts

7. The current DSA baseline reflects the latest macroeconomic developments and an active policy scenario under the ECF. It assumes a moderate fiscal consolidation driven mainly by a higher tax revenue following reforms in tax and customs administrations. It also reflects gradual improvements in the external situation due to the normalization of oil and other commodities prices:

- Growth is projected to rise to 3.0 percent in 2023 and then to an average of 4.1 percent in 2024-28, driven by large investment projects, including hospital expansion, the construction of a hotel complex, and the refurbishment of roads, while remittances are expected to decline from their earlier record levels during the COVID-19 pandemic amid lower growth in the eurozone, albeit remaining strong. In the long run, the boosting effect of those investment projects will dissipate, and growth will come back to an average of 3.8 percent. The growth path is similar to the one underlying the DSA for the ECF request; the impact on growth of the fiscal consolidation programmed under the ECF is expected to be muted due to the relatively small size of the public sector, the households' reliance on remittances, and the significant contribution of external grants to investment. Moreover, reforms under the DPOs and the ECF are expected to have a long-lasting effect while helping catalyze support from other development partners. The long-term growth projections are driven by a projected gradual increase in productivity and private capital accumulation as the country is projected to benefit from the completion of transport and energy projects, while financial intermediation is projected to improve and support private investment.

- Inflation will fall from an average of 12.4 percent in 2022 to 9.2 percent in 2023 (vs. 10.3 percent projected at the ECF request). After 2023, inflation will gradually stabilize at around 2 percent reflecting the peg of the exchange rate to the Euro.

- The fiscal sector assumptions reflect the policy adjustment scenario under the ECF program during 2023-27 built on the following:

    - An average increase in tax revenue of 0.3 percent of GDP per year from 2023 to 2027. Improvements will stem mainly from tax administration reforms aiming to raise the number of taxpayers, strengthen the use of the IT system, improve tax recovery from the SOEs and recover unpaid taxes. From the tax policy side, the authorities recently reintroduced the sales tax on construction material especially cement and end one-time subsidies and tax exemption introduced in 2022 as commodity prices stabilize.

    - The paring back of transfers and subsidies introduced in 2021-22 to mitigate the effects of the recent shocks, containing current spending while phasing out some one-offs (e.g., domestic outlays on the AU presidency and spending on elections); and reducing investment spending to pre-pandemic levels following the completion of some significant projects particularly El Maarouf Hospital.

- The domestic primary deficit is expected to improve from 1.9 percent of GDP in 2022 to 1.8 percent in 2023 and narrow further to 1.4 percent in 2024 reflecting the path of public investment spending, with fiscal consolidation (on domestic primary balance basis) during 2024-27 averaging 0.4 percent of GDP per year.

- Between 2023-2027, the overall fiscal consolidation is equivalent to 3 percent of GDP. It essentially reflects an increase in domestic revenue (1.3 percent) and a drop in domestic investment (2.4 percent). The rest stems from containing one-off current spending. The fiscal financing gap starting in 2023 mainly reflects the cost of the postal bank recapitalization, investments in the El Maarouf Hospital, and increased debt service obligations and lower revenues from key state-owned enterprises that need to recoup their earlier losses. Under the ECF, the gaps are expected to be filled by grants and concessional loans (see below). Outstanding arrears to external creditors of USD 8.2 million are assumed to be cleared over time. The long-term primary fiscal deficits are expected to average around 0.2 percent of GDP between 2033-2043, as fiscal adjustments and the mobilization of grant financing under the ECF reduce fiscal financing needs over the medium term and fiscal structural reforms sustainably increase revenues.

Text Table 3. Union of the Comoros: Factors Underpinning Fiscal Consolidation, 2023-27

Total revenues	-0.3
Domestic revenue	1.3
Tax administration	0.8
Customs Administration	0.4
Grants	-1.6
Primary Spending	-3.4
Transfers	-0.9
Investments	-2.4
Change in Primary deficit	-3.1

Sources: Comorian authorities, and IMF staff

- The external sector assumption worsened moderately compared to the DSA at the ECF request, reflecting strong imports. The external current account deficit is expected to be around 5.8 percent of GDP in 2023 (vs. 5.0 percent projected at the ECF request) before gradually declining to 5.3 percent in 2025. Exports of goods are projected to continue recovering while tourism has already been recording a stronger performance compared to pre-COVID-19 levels. Comoros would benefit from the projected increase in exports of tourism services as the quality of lodging services improves and the destination becomes more attractive with the implementation of the tourism sector strategy. Imports jumped to a high level amid soaring prices as a result of the war in Ukraine and will continue to be spurred by substantial public investment in 2023-24. Remittance inflows are expected to remain strong despite the correction from the COVID-19 levels, given the expected continued importance of the Comorian diaspora. With sustained import growth and increased interest payments from previously contracted non-concessional debt, the current account deficit is projected to remain at around 5½ percent of GDP over the next two years before settling at around 3.3 percent of GDP on average over the long term. Foreign exchange reserves coverage is projected to improve over the horizon while the fiscal financing gap translates into an external financing gap of the same size as the government faces tight domestic and external borrowing constraints.

8. This DSA assumes that the authorities will adopt a prudent borrowing strategy. The assumption is that the government mobilizes mostly external financing, with an average borrowing mix of 95 percent for external borrowing over the projected horizon.

- **Foreign financing**: Staff assumes that disbursement from already contracted loans will continue over the coming years in line with the authorities' plans. IDA financing, including through investment projects, is projected at USD 20.4 million per year on average during the period 2023-2026. The baseline includes around USD 50 million from AfDB, USD 12.8 million from Saudi Arabia over 3 years, and the Galawa Hotel loan. The baseline also includes around USD 20 from the ISDB for the financing of the El Maarouf Hospital with the remaining financing expected from other multilateral partners over the medium-term. For the purposes of this DSA, staff further assumes that the fiscal financing gap during 2023-26 would be filled with grants and concessional loans. These assumptions reflect financing under the ECF and DPO programs, which catalyzes multilateral and bilateral partners' financing, as well as Comoros' commitment to limiting non-concessional borrowing under the ECF and as a PPA under the SDFP.[4] The average grant element of new borrowing is 49.1 percent.

Text Table 4. Union of the Comoros: Summary Table of Projected External Borrowing Program

PPG external debt	2023		2024		2025		2026	
	USD million	Percent	USD million	Percent	USD million	Percent	USD million	Percent
By sources of debt financing	**116.7**	**100**	**79.2**	**100.0**	**81.0**	**100.0**	**65.4**	**100**
Concessional debt, of which	**64.1**	**54.9**	**69.7**	**88.0**	**71.5**	**88.3**	**55.9**	**85.5**
Multilateral debt	59.8	51.3	47.9	60.5	36.5	45.1	5.9	9.0
Bilateral debt	4.3	3.6	9.4	11.8	9.3	11.5	5.1	7.8
Other	0.0	0.0	12.5	15.8	25.7	31.7	44.9	68.7
Non-concessional debt, of which	**52.6**	**45.1**	**9.5**	**12.0**	**9.5**	**11.7**	**9.5**	**14.5**
Semi-concessional[1]	9.5	8.1	9.5	12.0	9.5	11.7	9.5	14.5
Commercial terms	43.1	36.9	0.0	0.0	0.0	0.0	0.0	0.0
By Creditor Type	**116.7**	**100.0**	**79.2**	**100.0**	**81.0**	**100.0**	**65.4**	**100.0**
Multilateral debt	69.3	59.4	57.4	72.4	46.0	56.8	15.4	23.5
Bilateral debt	4.3	3.6	9.4	11.8	9.3	11.5	5.1	7.8
Commercial terms	43.1	36.9	0.0	0.0	0.0	0.0	0.0	0.0
Other	0.0	0.0	12.5	15.8	25.7	31.7	44.9	68.7
		0.0		0.0		0.0		0.0
Uses of debt financing	**116.7**	**100.0**	**79.2**	**100.0**	**81.0**	**100.0**	**65.4**	**100.0**
Infrastructure	64.1	54.9	57.2	72.3	45.8	56.6	34.1	52.1
Budget Financing	9.5	8.1	9.5	12.0	9.5	11.7	9.5	14.5
Other	43.1	36.9	12.5	15.8	25.7	31.7	21.8	33.4

1/ Projected ECF disbursements

- **Domestic financing:** the increase of the domestic financing in 2023 reflects mainly disbursement in 2023 for ONICOR, the SOE in charge of the rice import. However, since the overall financial situation of the SOEs, altogether generally improved, the domestic debt trajectory, declines from an average (2023-2028) 6.3 percent of GDP (in the previous DSA) to 5.0 percent of GDP. Overall, domestic borrowing in this DSA reflects the rollover of SOE debt, which is assumed as medium-term debt at an average nominal interest rate of 6 percent.

[4] The ECF access is proposed at 180 percent of quota (SDR 32.04 million) disbursed over four years. The 2023 DPO is USD 20 million.

Text Table 5. Union of the Comoros: Assumptions for DSA

	2020	2021 Prel.	2022 Proj.	2023 Proj.	2024 Proj.	2025 Proj.	2026 Proj.	2027 Proj.	2028 Proj.	Long Term[1]
GDP Growth Rate (percent)										
Previous DSA	-0.2	2.1	2.6	3.0	3.5	4.0	4.3	4.5	4.3	3.8
Current DSA	-0.2	2.0	2.6	3.0	3.5	4.0	4.3	4.5	4.3	3.8
Inflation, average (percent)										
Previous DSA	0.8	0.0	12.4	10.3	2.0	2.1	2.0	1.9	1.9	1.9
Current DSA	0.8	0.0	12.4	9.2	2.2	2.3	2.2	1.9	1.9	1.9
Government Revenues (percent of GDP)										
Previous DSA	9.2	10.3	9.8	9.7	9.9	10.3	10.7	11.0	11.3	13.6
Current DSA	9.2	10.4	9.9	10.3	10.4	10.9	11.2	11.6	11.8	13.9
Overall Fiscal Balance (percent of GDP)										
Previous DSA	-1.4	-3.4	-1.9	-2.4	-1.8	-1.1	-0.2	0.0	-0.1	0.0
Current DSA	-1.4	-3.4	-1.9	-1.8	-1.4	-0.2	-0.2	-0.1	-0.1	0.0
FDI (percent of GDP)										
Previous DSA	0.3	0.3	0.4	0.4	0.5	0.6	0.6	0.6	0.6	0.6
Current DSA	0.3	0.3	0.3	0.4	0.5	0.6	0.6	0.6	0.6	0.6
Current Account Deficit (Percent of GDP)										
Previous DSA	-1.9	-0.5	-2.4	-5.0	-5.1	-4.5	-4.5	-4.3	-4.3	-4.3
Current DSA	-1.8	-0.3	-0.5	-5.8	-5.7	-5.3	-5.5	-5.2	-4.4	-3.7
Export of Goods and Services (annual percentage growth)										
Previous DSA	-48.3	62.4	76.7	-20.0	8.2	8.7	9.1	13.2	8.3	6.5
Current DSA	-44.7	58.7	73.7	-28.8	3.7	7.8	8.7	11.8	10.2	7.5
Import of Goods and Services (annual percentage growth)										
Previous DSA	-2.5	2.4	16.6	17.4	4.5	3.8	6.7	6.7	6.3	6.8
Current DSA	-2.5	2.4	20.3	11.9	3.1	3.0	8.5	6.8	6.7	6.4
Investment (percent of GDP)										
Previous DSA	15.6	15.7	16.7	17.3	17.6	17.7	17.9	18.0	18.1	18.7
Current DSA	15.6	15.9	16.9	17.5	17.8	17.9	18.1	18.2	18.3	18.9
Net Current transfers (percent of GDP)										
Previous DSA	20.2	21.6	21.3	20.4	18.6	18.3	17.6	17.0	16.5	15.1
Current DSA	20.2	21.8	24.7	19.7	19.0	17.1	16.4	16.2	16.5	15.1

1 Covers the period 2029-42 for current DSA, and 2029-41 for the previous DSA.

Source: IMF staff

9. Realism tools do not flag issues (Figures 3 and 4). The 3-year adjustment in the primary balance puts Comoros in the mid-range of adjustments seen in other LIC program countries. The projected GDP growth in 2023 is compatible with a range of plausible fiscal multipliers. The growth forecast for 2024 exceeds the fiscal multiplier expectation due to positive spillover effects from the ECF program's structural reforms and the anticipated recovery in global demand conditions which is expected to boost exports. The projected contribution of government investment over the medium term is in line with historical experience. The projected contribution from the other factors reflects the recovery from the multiple successive shocks during 2019-22, and the gradual increase in private sector and export activities, reflecting improvements in the business climate due to reforms under the ECF. The residual shown in the public debt dynamics for 2023 (Table 3) captures the higher uptake of commercial bank loans by SOEs and the Galawa loan.[5] Similarly, most of the medium-term decline in debt is explained by the residual that captures repayments of the Galawa hotel loan as well as the Comores Telecom loan. Most of the unexpected changes in the public debt ratio over the past five years reflected the expansion in the debt coverage. Trends in debt indicators under the historical scenarios (both for public external and overall public debt) mimic those of the baseline scenario providing credibility to the baseline macroeconomic scenario.

C. Country Classification and Determination of Stress Test Scenarios

10. Comoros is assessed to have medium debt carrying capacity, unchanged from the May 2023 DSA (Text Table 6), with a Composite Indicator (CI) of 2.82 based on October 2023 WEO

[5] The Galawa loan is a balance of payment transaction and therefore not captured by the primary deficit variable in Table 3.

projections and the 2022 Country Policy and Institutional Assessment (CPIA) rating. As a result, this DSA continues to use the same external debt burden thresholds and total public benchmarks for countries with medium debt carrying capacity (Text Table 6). The relevant thresholds indicating high risk are 40 percent for the PV of debt-to-GDP ratio, 180 percent for the PV of debt-to-exports ratio, 15 percent for the debt service-to-exports ratio, and 18 percent for the debt service-to revenue ratio. These thresholds are applicable to public and publicly guaranteed external debt. For the PV of total public debt-to-GDP ratio, the benchmark is 55 percent.

11. Standardized stress tests use default settings. In addition, Comoros qualifies for the natural disaster scenario given that it is exposed and vulnerable to natural disaster shocks, such as cyclones and eruptions of the Karthala volcano. The tailored stress test assumes a one-off natural disaster shock of 10 percent of GDP to external PPG debt-GDP ratio in the second year of the projection period. The financing cost of all shocks is set to the default values of the models.

Text Table 6. Union of the Comoros: Calculation and Contribution of the CI Score Components

Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintages
Medium	Medium 2.82	Medium 2.83	Medium 2.85

Applicable Thresholds

External debt burden thresholds	
PV of debt in % of	
Exports	180
GDP	40
Debt service in % of	
Exports	15
Revenue	18

Total public debt benchmark	
PV of total public debt in percent of GDP	55

Source: LIC DSF

DEBT SUSTAINABILITY

A. External Debt Sustainability

12. Three out of four external debt burden indicators breach their respective thresholds under the baseline macroeconomic assumptions underlying this exercise (Figure 1). As in the DSA for the ECF request, breaches are temporary and not of significant importance. Compared with the ECF request DSA, the breaches are smaller reflecting the reduced financing needs from the stronger-than-expected fiscal consolidation and the lower estimated cost of completing the El Maarouf Hospital construction. Results of the external debt sustainability assessment can be summarized as follows:

- The PV of external debt-to-GDP ratio is still below the threshold and is expected to decline along the same trajectory of the ECF request DSA. Similarly, the breach of the PV of debt-to-export ratio persists during 2023-25 before declining below the threshold.

- The debt service-to-exports ratio and the debt service-to-revenue ratio continue to breach their respective thresholds. However, the magnitude of the breaches during 2024-28, identified in the previous DSA for the ECF request, have been mitigated due to the stronger-than-expected revenue mobilization in 2023H1 as well as an improvement in the external environment.

13. The stress tests suggest that the export shock has the largest negative impact on the debt trajectory. The export shock would cause threshold breaches for three of four indicators (PV of debt-to-export, debt service-to-export, and debt service-to-revenue). The PV of debt-to-export ratio is the most sensitive indicator as it would see prolonged breaches of its threshold under most of the shock scenarios.

B. Total Public Debt Sustainability

14. Overall public debt is projected to peak by 2024 before declining progressively, remaining well below its indicative benchmark under the baseline and stress tests (Figure 2). Under the current baseline, domestic debt accumulation would continue to decrease due to the fiscal space resulting from the proposed fiscal consolidation and the moderate improvement of the financial situation of SOEs from less adverse international commodity prices. The results of the stress tests indicate that the most significant negative effects on the debt trajectory are caused by the other flows[6] shock. The other flows shock would impact the present value of the debt-to-revenue ratio by 22 percent in the second period after the shock and it will persist over the projection period (2024-2043) . The most significant shocks on the debt service-to-revenue ratio are the depreciation and the other flows shocks. The former is mitigated by Comoros' fixed exchange rate while the latter would increase debt service-to-revenue ratio by an average of 1.1 percentage points over 2024-2043.

C. Risk Assessment and Vulnerabilities

15. Comoros external and overall debt are sustainable but remain at a high risk of debt distress. Both ratings on the external public debt and overall public debt are determined by the levels and trajectories of debt, together with any breaches of the debt indicators. The levels and trajectories continue to be assessed as sustainable. The magnitude of the breaches of the external debt indicators is limited and the projected debt burden metrics have improved over the forecast horizon as a result of the envisaged fiscal consolidation. Additional considerations that mitigate the mechanical risk signals in support of the analysis include the authorities' economic reform program supported by the ECF and the World Bank's DPOs, which are expected to catalyze grant and concessional financing. Another factor is the consistently strong performance of remittances, which supports external sustainability. This overall assessment is contingent on: (i) progress on domestic resource mobilization and gradual but sustained fiscal consolidation over the medium term, (ii) no additional non-concessional borrowing, and (iii) enhanced financial oversight of SOEs.

[6] Includes official and private transfers and FDI.

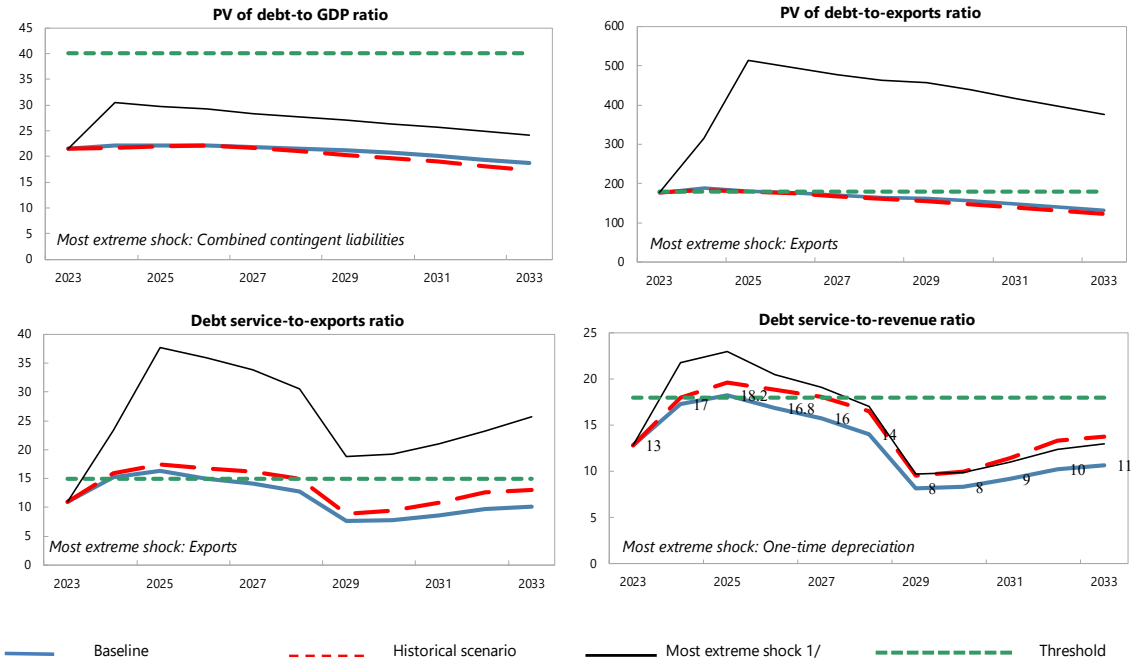
16. Risks to the baseline are tilted to the downside. Delays in implementing the needed fiscal adjustment and structural reforms, compounded with delays in the needed external financing and deterioration in global conditions may lead to a weakening of the macroeconomic situation. Contracting of further debt, particularly non-concessional, would elevate debt sustainability risks.

17. Mitigating risks to public debt requires several policy actions: Reforms under the ECF program that will help mitigate risks to public debt via (i) making faster progress on domestic resource mobilization and ensuring gradual but sustained fiscal consolidation over the medium term, (ii) avoiding additional non-concessional borrowing, (iii) enhancing financial oversight of SOEs, which should help reduce the size of a contingent liability shock should it materialize. Risks could also be mitigated through the implementation of reforms under the World Bank's DPOs and the SDFP as follows: (i) the compulsory reporting of SOE financial performance and the audit of financial statements; and (ii) the enhanced regulatory framework for the issuance of public guarantees and on-lend credit. Additional measures related to the diversification of the export base could help further lessen debt vulnerabilities as underscored by the sensitivity analysis.

Authorities' Views

18. The authorities generally shared staff's assessment of debt sustainability and reaffirmed their commitment to continuing the strong implementation of the IMF-supported program. They reiterated that fiscal consolidation, supported by fiscal structural reforms, will be needed to preserve debt sustainability, and gradually reduce the risk of debt distress to moderate over the medium term. The authorities are committed to prudent borrowing policies and noted that budget support, including that from the IMF and the WB, has been critical for containing borrowing needs.

Figure 1. Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2023-2033



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	Yes	
Natural disaster	No	No
Commodity price 2/	n.a.	n.a.
Market financing	n.a.	n.a.

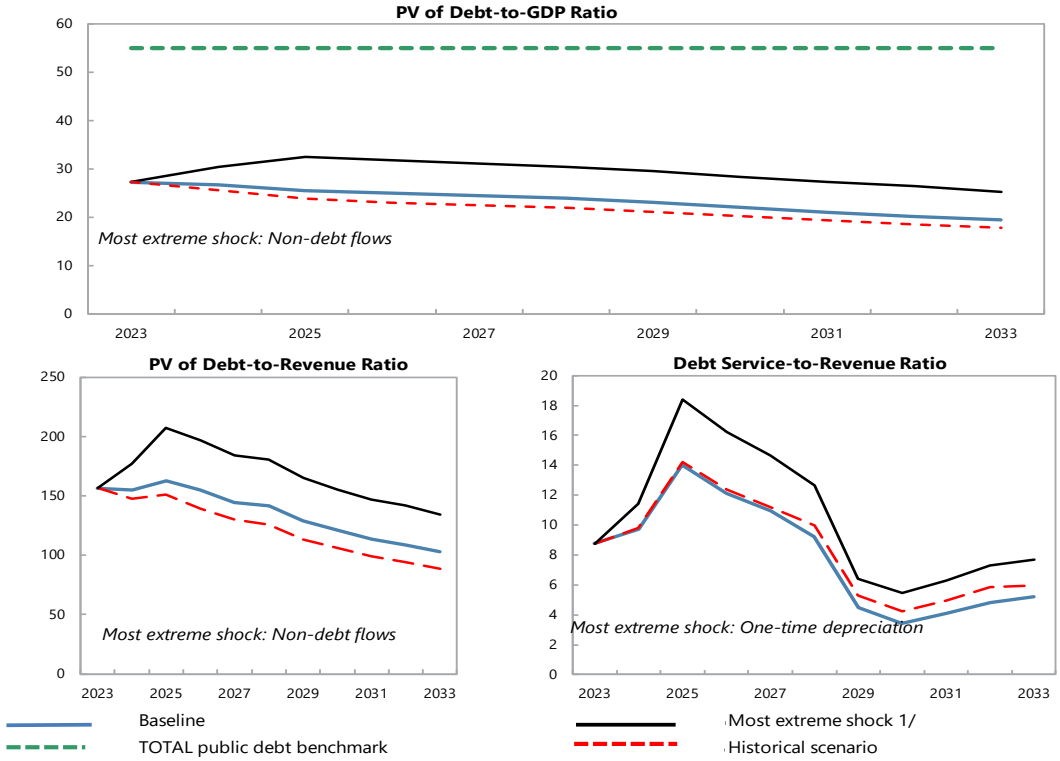
Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.1%	1.1%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	33	33
Avg. grace period	8	8

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities and staff estimates and projections.
1/ The most extreme stress test is the test that yield the highest ratio in or before 2033. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most extreme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.
2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF-research department.

Figure 2. Indicators of Public Debt Under Alternative Scenarios, 2023-2033



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	130%	130%
Domestic medium and long-term	-30%	-30%
Domestic short-term	-1%	-1%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.1%	1.1%
Avg. maturity (incl. grace period)	33	33
Avg. grace period	8	8
Domestic MLT debt		
Avg. real interest rate on new borrowing	4.0%	4.0%
Avg. maturity (incl. grace period)	6	6
Avg. grace period	0	0
Domestic short-term debt		
Avg. real interest rate	-3.3%	-3.3%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

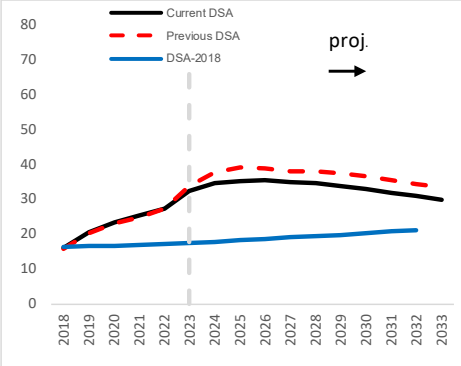
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2033. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

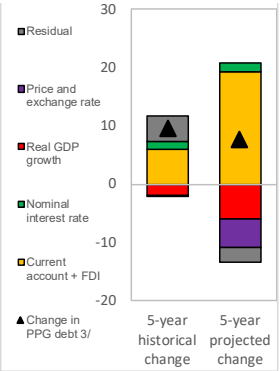
Figure 3. Drivers of Debt Dynamics – Baseline Scenario External Debt

External Debt

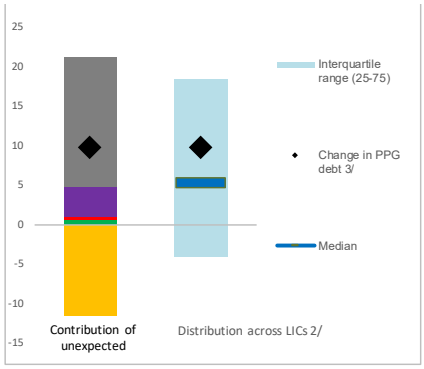
Gross Nominal PPG External Debt
(In percent of GDP, DSA vintages)



Debt-creating Flows
(Percent of GDP)

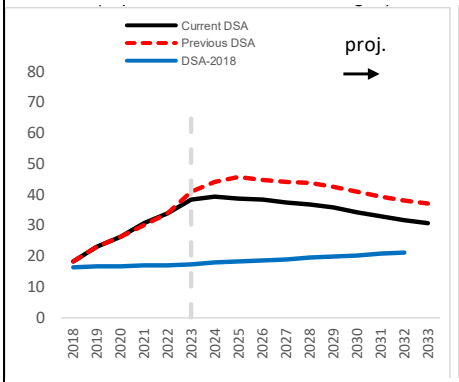


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

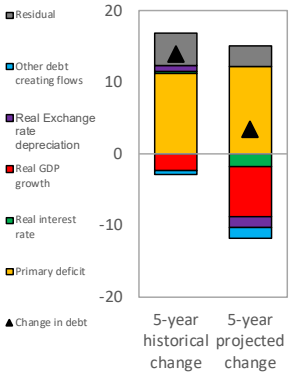


Public Debt

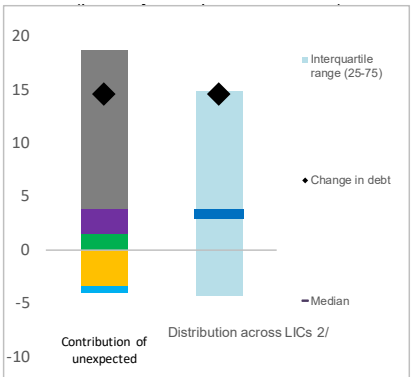
Gross Nominal PPG External Debt
(In percent of GDP, DSA vintages)



Debt-creating Flows
(Percent of GDP)



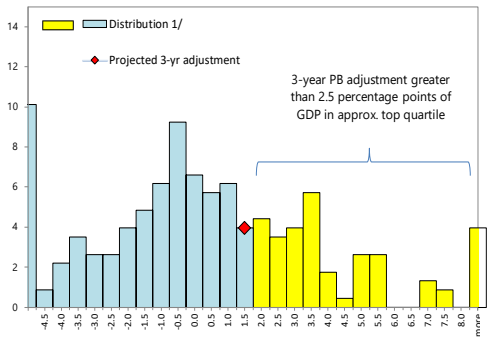
Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



Sources: Country authorities; and staff.
1/ Difference between anticipated and actual contributions on debt ratios.
2/ Distribution across LICs for which LIC DSAs were producted.
3/ Given the relatively low priavte external debt for average lowincome countries, a ppt change in PPG external debt should be largely explained by the drivers of the eternal dynamic equation.

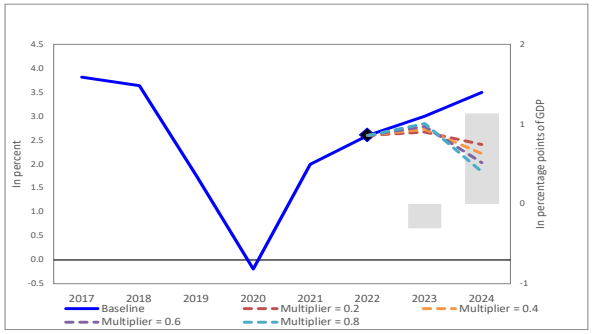
Figure 4. Realism Tools

3-year Adjustment in Primary Balance
(Percentage Points of GDP)



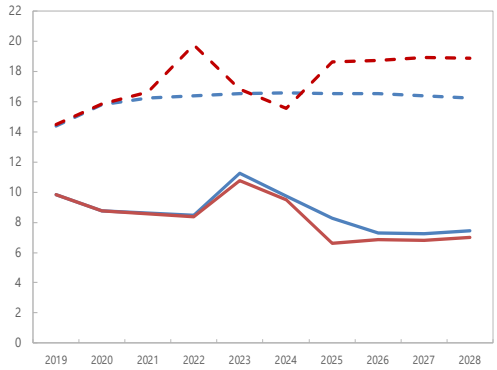
1/ Date cover Fund-supported programs for LICS (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths/1

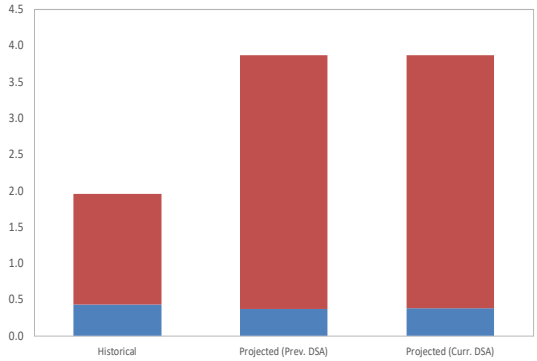


1/ Bars refer to annual projected fiscal adjustment (right-hand scale), and lines show possible real GDP growth paths under fiscal multipliers (left-hand scale).

Public and Private Investments Rates
(Percent of GDP)



Contribution to Real GDP Growth
(Percent, 5-year average)



Sources: Country authorities; and staff estimates and projections.
The significant difference between 'Prev. DSA' and 'Curr. DSA' lines in the 'public and private investment rates' chat is predominaly due to calculation error in the previous DSA. Specifically, foreign foreign financed public investment was erroneously included in prvate investment rather than public investment, hence artificially inflating the former and deflating the latter. The sum of private and public investment in the two DSA versions is boradly similar.

Table 1. Union of the Comoros: Decomposition of Public Debt 2020-22[1]

	2020	2021	2022	2020	2021	2022
	Million USD			% GDP		
Total	320.0	388.8	416.3	26.3	30.5	34.0
External	284.5	322.5	333.5	23.4	25.3	27.2
Multilateral creditors[2]	106.4	123.0	131.1	8.7	9.7	10.7
IMF	28.7	27.6	23.4	2.4	2.2	1.9
World Bank	20.2	28.4	39.2	1.7	2.2	3.2
ADB/AfDB/IADB	2.5	7.1	8.2	0.2	0.6	0.7
Other Multilaterals	55.1	60.0	60.2	4.5	4.7	4.9
o/w: BADEA	28.3	28.0	28.0	2.3	2.2	2.3
o/w: TDB	13.2	18.3	17.4	1.1	1.4	1.4
o/w: IsDB	7.5	7.5	6.9	0.6	0.6	0.6
Bilateral Creditors	178.1	199.4	202.5	14.6	15.7	16.5
Paris Club	2.1	2.1	2.5	0.2	0.2	0.2
o/w: France	2.1	2.1	2.5	0.2	0.2	0.2
Non-Paris Club	176.0	197.4	200.0	14.5	15.5	16.3
o/w: China	91.1	112.1	110.8	7.5	8.8	9.0
o/w: Saudi Arabia	33.7	33.7	38.0	2.8	2.6	3.1
o/w: India	26.4	26.4	26.4	2.2	2.1	2.2
o/w: Kuwait	24.8	25.1	24.8	2.0	2.0	2.0
Bonds	0.0	0.0	0.0	0.0	0.0	0.0
Commercial creditors	0.0	0.0	0.0	0.0	0.0	0.0
Domestic	35.5	66.3	82.7	2.9	5.2	6.8
Held by residents, total[3]	35.5	66.3	82.7	2.9	5.2	6.8
SOEs debt[4]	N.A	23.0	19.2	N.A	1.8	1.6
Memo items:						
Collateralized debt[5]	0.0	0.0	0.0	0.0	0.0	0.0
o/w: Related	0.0	0.0	0.0	0.0	0.0	0.0
o/w: Unrelated	0.0	0.0	0.0	0.0	0.0	0.0
Contingent liabilities[6]	107.2	159.1	153.1	8.8	12.5	12.5
o/w: Public guarantees	0.0	0.0	0.0	0.0	0.0	0.0
o/w: Other explicit contingent liabilities[7]	107.2	159.1	153.1	8.8	12.5	12.5
External arrears	N.A	6.0	7.2	N.A	0.5	0.6
Nominal GDP	1218	1273	1225			

Source: Country authorities and staff estimates and projections

1/ As reported by Country authorities according to their classification of creditors, including by official and commercial.

2/ "Multilateral creditors" are simply institutions with more than one official shareholder and may not necessarily align with creditor classification under other IMF policies (e.g. Lending Into Arrears).

3/ Inludes central bank on lending related to the SDR allocation.

4/ SOEs debt in Comoros is guarantee debt.

5/ Debt is collateralized when the creditor has rights over an asset or revenue stream that would allow it, if the borrower defaults on its payment obligations, to rely on the asset or revenue stream to secure repayment of the debt.

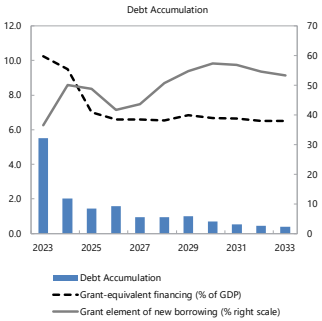
6/ As defined in the customized contingent liabilities stress test of the last three published DSAs.

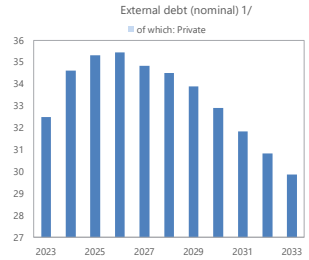
7/ Includes other-one off guarantees not included in publicly guaranteed debt (e.g. credit lines) and other explicit contingent liabilities not elsewhere classified (e.g. potential legal claims, payments resulting from PPP arrangements).

Table 2. Union of the Comoros: External Debt Sustainability Framework, Baseline Scenario, 2020-2043

	Actual			Projections												Average 8/	
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2043	Historical	Projections
External debt (nominal) 1/	23.5	25.4	27.3	32.5	34.6	35.3	35.4	34.8	34.5	33.9	32.9	31.8	30.8	29.9	20.6	17.9	33.3
of which: public and publicly guaranteed (PPG)	23.5	25.4	27.3	32.5	34.6	35.3	35.4	34.8	34.5	33.9	32.9	31.8	30.8	29.9	20.6	17.9	33.3
Change in external debt	3.0	2.0	1.9	5.2	2.1	0.7	0.1	-0.6	-0.3	-0.6	-1.0	-1.1	-1.0	-1.0	-0.9		
Identified net debt-creating flows	1.0	-1.0	1.2	4.7	3.6	2.0	2.3	2.2	2.3	3.0	2.3	2.0	2.2	2.1	1.7	1.5	2.6
Non-interest current account deficit	1.6	0.1	-0.1	5.5	4.8	3.5	4.0	4.0	4.1	4.6	3.8	3.5	3.7	3.6	2.9	2.2	4.1
Deficit in balance of goods and services	22.5	22.6	25.7	25.8	24.8	22.6	22.2	21.6	21.1	20.8	19.9	19.7	19.3	19.1	18.3	19.5	21.5
Exports	5.7	10.1	13.6	12.1	11.7	12.2	12.6	12.8	13.1	13.1	13.3	13.6	13.8	14.1	16.1		
Imports	28.2	32.8	39.3	37.9	36.5	34.8	34.8	34.4	34.2	33.9	33.2	33.3	33.1	33.2	34.4		
Net current transfers (negative = inflow)	-20.2	-21.8	-24.7	-19.7	-19.5	-18.5	-17.6	-17.0	-16.5	-15.7	-15.7	-15.7	-15.1	-15.1	-14.9	-16.7	-16.9
of which: official	-3.4	-1.8	-3.2	-1.9	-3.0	-2.1	-1.3	-1.3	-1.2	-1.9	-1.9	-1.9	-1.7	-1.7	-1.5		
Other current account flows (negative = net inflow)	-0.7	-0.8	-1.1	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.6	-0.6	-0.5
Net FDI (negative = inflow)	-0.3	-0.3	-0.3	-0.4	-0.5	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.4	-0.6
Endogenous debt dynamics 2/	-0.3	-0.8	1.6	-0.4	-0.6	-0.9	-1.2	-1.2	-1.2	-1.0	-0.9	-0.9	-0.8	-0.8	-0.6		
Contribution from nominal interest rate	0.2	0.3	0.6	0.3	0.4	0.4	0.3	0.3	0.2	0.3	0.3	0.3	0.3	0.3	0.2		
Contribution from real GDP growth	0.0	-0.4	-0.7	-0.7	-1.1	-1.3	-1.4	-1.5	-1.4	-1.2	-1.2	-1.2	-1.1	-1.1	-0.8		
Contribution from price and exchange rate changes	-0.5	-0.6	1.7	...	...	...	...	...	...	...	...	...	...	...	...		
Residual 3/	2.1	3.0	0.6	0.5	-1.5	-1.3	-2.1	-2.8	-2.6	-3.7	-3.2	-3.1	-3.2	-3.1	-2.6	-1.2	-2.4
of which: exceptional financing	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators																	
PV of PPG external debt-to-GDP ratio	...	...	18.1	21.5	22.1	22.1	22.1	21.8	21.4	21.2	20.7	20.0	19.4	18.7	12.4		
PV of PPG external debt-to-exports ratio	...	...	132.9	177.3	188.2	181.4	175.8	169.5	164.3	162.1	155.5	147.6	139.9	132.3	77.3		
PPG debt service-to-exports ratio	12.3	7.3	8.9	10.9	15.3	16.3	15.0	14.2	12.7	7.6	7.8	8.6	9.6	10.1	5.6		
PPG debt service-to-revenue ratio	7.4	7.1	12.4	12.8	17.3	18.2	16.8	15.7	14.0	8.2	8.3	9.1	10.2	10.7	5.6		
Gross external financing need (Million of U.S. dollars)	24.1	6.1	10.0	87.1	86.8	75.2	86.7	90.1	94.6	97.4	86.2	88.7	101.5	106.5	135.2		
Key macroeconomic assumptions																	
Real GDP growth (in percent)	-0.2	2.0	2.6	3.0	3.5	4.0	4.3	4.5	4.3	3.8	3.8	3.8	3.8	3.8	3.8	2.5	3.9
GDP deflator in US dollar terms (change in percent)	2.7	2.5	-6.2	6.7	2.6	2.6	2.3	1.6	1.5	2.0	2.0	2.0	2.0	2.0	2.0	-0.3	2.5
Effective interest rate (percent) 4/	1.1	1.2	2.3	1.3	1.4	1.1	0.8	0.8	0.7	0.8	1.0	1.0	1.0	1.0	1.1	0.8	1.0
Growth of exports of G&S (US dollar terms, in percent)	-51.1	87.2	28.8	-2.2	3.1	10.6	10.6	8.2	7.6	6.1	7.6	8.0	7.9	7.8	6.0	12.1	6.8
Growth of imports of G&S (US dollar terms, in percent)	-2.9	21.4	15.4	5.8	2.5	1.7	6.6	5.0	5.2	5.0	3.8	5.9	5.3	6.2	5.8	5.0	4.8
Grant element of new public sector borrowing (in percent)	...	...	...	36.4	50.1	48.8	41.7	43.6	50.8	54.8	57.3	56.9	54.6	53.3	53.9	...	49.8
Government revenues (excluding grants, in percent of GDP)	9.4	10.4	9.7	10.3	10.4	10.9	11.2	11.6	11.8	12.2	12.5	12.8	13.0	13.3	16.1	10.1	11.8
Aid flows (in Million of US dollars) 5/	107.7	85.9	56.0	143.3	143.1	111.1	131.1	137.9	147.0	150.2	151.2	159.4	168.4	179.5	277.8		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	10.2	9.5	7.0	6.6	6.6	6.5	6.8	6.7	6.7	6.5	6.5	6.0	...	7.2
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	65.0	78.0	75.3	73.8	79.7	81.3	88.1	90.6	90.6	88.9	88.2	93.0	...	81.8
Nominal GDP (Million of US dollars)	1,218	1,273	1,225	1,346	1,429	1,525	1,629	1,728	1,830	1,937	2,051	2,171	2,297	2,431	4,287		
Nominal dollar GDP growth	2.5	4.5	-3.8	9.9	6.2	6.7	6.8	6.1	5.9	5.8	5.9	5.9	5.8	5.8	5.8	2.1	6.4
Memorandum items:																	
PV of external debt 7/	...	...	18.1	21.5	22.1	22.1	22.1	21.8	21.4	21.2	20.7	20.0	19.4	18.7	12.4		
In percent of exports	...	...	132.9	177.3	188.2	181.4	175.8	169.5	164.3	162.1	155.5	147.6	139.9	132.3	77.3		
Total external debt service-to-exports ratio	12.3	7.3	8.9	10.9	15.3	16.3	15.0	14.2	12.7	7.6	7.8	8.6	9.6	10.1	5.6		
PV of PPG external debt (in Million of US dollars)			221.2	288.7	315.8	336.5	360.7	376.3	392.4	410.6	424.1	434.8	444.7	453.4	532.2		
(PVt-PVt-1)/GDPt-1 (in percent)				5.5	2.0	1.4	1.6	1.0	0.9	1.0	0.7	0.5	0.5	0.4	0.2		
Non-interest current account deficit that stabilizes debt ratio	-1.5	-1.9	-1.9	0.3	2.7	2.8	3.9	4.6	4.4	5.2	4.7	4.6	4.7	4.5	3.7		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα(1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

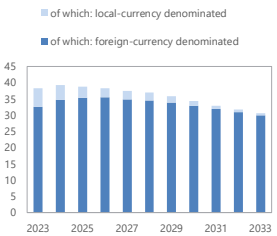
8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

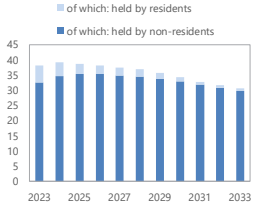
Table 3. Union of the Comoros: Public Sector Debt Sustainability Framework, Baseline Scenario, 2020-2043

	Actual			Projections								Average 6/	
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2033	2043	Historical	Projections
Public sector debt 1/	**26.4**	**30.6**	**34.0**	**38.2**	**39.2**	**38.7**	**38.2**	**37.4**	**36.9**	**30.6**	**20.0**	**20.5**	**35.8**
of which: external debt	**23.5**	**25.4**	**27.3**	**32.5**	**34.6**	**35.3**	**35.4**	**34.8**	**34.5**	**29.9**	**20.6**	**17.9**	**33.3**
Change in public sector debt	3.3	4.3	3.4	4.2	0.9	-0.4	-0.5	-0.8	-0.5	-1.1	-1.0		
Identified debt-creating flows	**-0.3**	**1.7**	**4.7**	**2.4**	**1.1**	**-0.2**	**-0.3**	**-0.9**	**-0.6**	**-1.0**	**-0.5**	**-0.3**	**-0.4**
Primary deficit	**0.0**	**2.5**	**3.8**	**4.1**	**2.9**	**2.1**	**2.0**	**1.2**	**1.4**	**0.5**	**0.1**	**0.1**	**1.4**
Revenue and grants	18.3	17.1	14.3	17.4	17.2	15.7	16.1	16.8	16.8	18.9	21.6	**17.7**	**17.5**
of which: grants	8.8	6.7	4.6	7.1	6.8	4.8	4.9	5.3	5.0	5.5	5.5		
Primary (noninterest) expenditure	18.2	19.6	18.1	21.5	20.1	17.8	18.1	18.0	18.2	19.3	21.7	**17.8**	**18.9**
Automatic debt dynamics	**-0.2**	**-0.7**	**1.4**	**-1.2**	**-1.5**	**-1.9**	**-2.1**	**-2.1**	**-2.0**	**-1.5**	**-0.6**		
Contribution from interest rate/growth differential	0.2	-0.7	-0.7	-1.2	-1.5	-1.9	-2.1	-2.1	-2.0	-1.5	-0.6		
of which: contribution from average real interest rate	0.1	-0.2	0.1	-0.2	-0.2	-0.4	-0.4	-0.4	-0.5	-0.4	0.1		
of which: contribution from real GDP growth	0.0	-0.5	-0.8	-1.0	-1.3	-1.5	-1.6	-1.6	-1.5	-1.2	-0.8		
Contribution from real exchange rate depreciation	-0.4	0.0	2.1	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**-0.1**	**-0.4**	**-0.5**	**-0.3**	**-0.4**	**-0.2**	**0.0**	**0.0**	**0.0**	**0.0**	**-0.1**	**-0.1**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	-0.1	-0.4	-0.5	-0.3	-0.4	-0.2	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**3.6**	**2.6**	**-1.3**	**1.9**	**-0.2**	**-0.2**	**-0.2**	**0.1**	**0.1**	**0.0**	**-0.4**	**1.3**	**0.1**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**24.8**	**27.2**	**26.7**	**25.5**	**24.9**	**24.4**	**23.9**	**19.4**	**11.9**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**173.5**	**156.6**	**155.1**	**162.9**	**154.6**	**144.6**	**141.9**	**102.8**	**54.9**		
Debt service-to-revenue and grants ratio 3/	**4.2**	**4.8**	**9.6**	**8.8**	**9.7**	**14.0**	**12.1**	**10.9**	**9.2**	**5.2**	**1.4**		
Gross financing need 4/	0.7	3.2	4.7	5.1	4.3	3.8	3.7	3.0	2.9	1.4	0.4		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	-0.2	2.0	2.6	3.0	3.5	4.0	4.3	4.5	4.3	3.8	3.8	**2.5**	**3.9**
Average nominal interest rate on external debt (in percent)	1.1	1.2	2.3	1.3	1.4	1.1	0.8	0.8	0.7	1.0	1.1	**0.8**	**1.0**
Average real interest rate on domestic debt (in percent)	1.7	4.2	-2.4	-1.3	-0.5	-1.6	-2.3	-2.5	-1.9	-10.4	18.4	**0.8**	**-3.8**
Real exchange rate depreciation (in percent, + indicates depreciation)	-1.9	0.0	8.4	...	...	...	...	...	...	...	...	**1.6**	**...**
Inflation rate (GDP deflator, in percent)	0.8	-1.2	5.3	3.3	2.1	2.3	2.2	2.0	1.9	2.0	2.0	**1.5**	**2.1**
Growth of real primary spending (deflated by GDP deflator, in percent)	-6.6	9.9	-5.6	22.2	-2.9	-8.1	6.2	4.0	5.6	5.3	4.5	**5.1**	**4.7**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-3.3	-1.7	0.4	-0.1	2.0	2.5	2.5	2.0	1.9	1.6	1.0	**-1.6**	**1.7**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, central bank, government-guaranteed debt . Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 4. Union of the Comoros: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2023-2033

	Projections 1/										
	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
PV of debt-to GDP ratio											
Baseline	20	21	21	21	21	20	20	20	19	19	18
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	20	20	19	19	18	17	16	16	15	14	13
B. Bound Tests											
B1. Real GDP growth	20	21	23	23	22	22	22	22	21	20	20
B2. Primary balance	20	23	25	25	24	24	24	23	23	22	21
B3. Exports	20	23	27	27	27	27	27	26	25	25	24
B4. Other flows 3/	20	24	27	27	26	26	26	25	25	24	23
B5. Depreciation	20	26	22	22	22	21	21	21	20	20	19
B6. Combination of B1-B5	20	26	28	28	27	27	27	26	26	25	24
C. Tailored Tests											
C1. Combined contingent liabilities	20	27	27	26	26	26	25	25	24	24	23
C2. Natural disaster	20	26	26	26	25	25	25	25	24	24	23
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	165	176	170	166	160	156	155	149	142	135	128
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	165	169	159	150	140	133	126	118	110	101	93
B. Bound Tests											
B1. Real GDP growth	165	176	170	166	160	156	155	149	142	135	128
B2. Primary balance	165	**193**	**205**	**197**	**190**	**185**	**183**	176	168	160	152
B3. Exports	165	**297**	**489**	**472**	**457**	**444**	**439**	**423**	**404**	**385**	**365**
B4. Other flows 3/	165	**206**	**219**	**212**	**205**	**199**	**197**	**190**	**181**	173	163
B5. Depreciation	165	176	141	138	134	130	130	125	118	112	106
B6. Combination of B1-B5	165	**254**	**204**	**285**	**276**	**268**	**265**	**256**	**244**	**232**	**219**
C. Tailored Tests											
C1. Combined contingent liabilities	165	**230**	**219**	**210**	**203**	**197**	**195**	**187**	179	171	163
C2. Natural disaster	165	**226**	**216**	**208**	**202**	**197**	**196**	**189**	**181**	174	166
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	11	15	**16**	15	13	12	7	7	8	9	10
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	11	**15**	**17**	**16**	**15**	14	8	8	10	11	12
B. Bound Tests											
B1. Real GDP growth	11	15	**16**	15	13	12	7	7	8	9	10
B2. Primary balance	11	15	**16**	**15**	14	13	7	8	9	9	11
B3. Exports	11	**22**	**37**	**35**	**32**	**29**	**17**	**18**	**20**	**22**	**25**
B4. Other flows 3/	11	15	**16**	**15**	14	13	8	8	9	10	11
B5. Depreciation	11	15	**16**	14	13	12	6	7	8	9	9
B6. Combination of B1-B5	11	**18**	**24**	**22**	**20**	**18**	11	11	12	14	**16**
C. Tailored Tests											
C1. Combined contingent liabilities	11	15	**17**	**15**	14	13	8	8	9	10	10
C2. Natural disaster	11	**15**	**17**	**16**	14	13	8	8	9	10	10
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	13	17	18	16	15	13	7	8	9	10	10
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	13	17	**19**	**18**	17	15	8	9	10	12	12
B. Bound Tests											
B1. Real GDP growth	13	17	**19**	18	16	14	8	8	9	11	11
B2. Primary balance	13	17	**18**	17	15	14	8	8	9	10	11
B3. Exports	13	17	**19**	**18**	16	15	9	9	10	11	12
B4. Other flows 3/	13	17	**18**	17	16	14	8	8	9	10	12
B5. Depreciation	13	**21**	**22**	**20**	18	16	9	9	10	12	12
B6. Combination of B1-B5	13	18	**21**	**19**	17	15	9	9	10	11	13
C. Tailored Tests											
C1. Combined contingent liabilities	13	17	**19**	17	16	14	8	8	9	10	11
C2. Natural disaster	13	17	**19**	17	16	14	8	8	9	10	11
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 5. Union of the Comoros: Sensitivity Analysis for Key Indicators of Public Debt, 2023-2033

	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
PV of Debt-to-GDP Ratio											
Baseline	27	27	26	25	24	24	23	22	21	20	19
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	27	26	24	23	22	22	21	20	19	19	18
B. Bound Tests											
B1. Real GDP growth	27	28	28	28	28	28	27	27	26	26	25
B2. Primary balance	27	28	29	28	28	28	27	26	25	24	23
B3. Exports	27	29	31	30	30	29	28	27	26	25	24
B4. Other flows 3/	27	30	32	32	31	30	29	28	27	26	25
B5. Depreciation	27	31	29	27	25	23	22	20	19	17	16
B6. Combination of B1-B5	27	28	28	24	24	23	22	21	21	20	19
C. Tailored Tests											
C1. Combined contingent liabilities	27	31	30	30	30	29	29	28	26	26	25
C2. Natural disaster	27	31	30	29	29	29	28	27	26	25	25
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
TOTAL public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	157	155	163	155	145	142	129	121	113	109	103
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	157	148	151	139	130	126	113	106	99	94	88
B. Bound Tests											
B1. Real GDP growth	157	159	175	169	160	160	148	142	137	135	131
B2. Primary balance	157	165	184	176	166	164	151	143	135	130	122
B3. Exports	157	167	198	188	176	173	157	148	140	135	128
B4. Other flows 3/	157	177	207	197	184	181	165	156	147	142	134
B5. Depreciation	157	189	188	170	153	143	126	115	104	95	85
B6. Combination of B1-B5	157	166	176	149	140	138	126	118	111	107	101
C. Tailored Tests											
C1. Combined contingent liabilities	157	181	194	186	176	174	160	152	143	137	130
C2. Natural disaster	157	178	190	182	172	171	157	149	141	136	130
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	9	10	14	12	11	9	4	3	4	5	5
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	9	10	14	12	11	10	5	4	5	6	6
B. Bound Tests											
B1. Real GDP growth	9	10	15	12	11	9	4	3	3	4	5
B2. Primary balance	9	10	12	9	9	7	3	2	4	6	6
B3. Exports	9	10	14	13	12	10	5	4	5	5	6
B4. Other flows 3/	9	10	14	13	12	10	5	4	5	5	7
B5. Depreciation	9	11	18	16	15	13	6	5	6	7	8
B6. Combination of B1-B5	9	10	14	11	10	8	4	3	4	5	5
C. Tailored Tests											
C1. Combined contingent liabilities	9	10	9	9	8	7	3	2	5	6	6
C2. Natural disaster	9	10	10	10	9	7	3	2	5	5	6
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.